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Intel Comes Out Swinging at Sun

Benchmarks pit Itanium 2 chips against UltraSparc III servers

BY TOM MURPHY

Intel Corp. has taken another shot at Sun Microsystems's dominance in servers with its new Itanium 2 chip, and this time it looks like it's closer to a bull's-eye. But Intel has to be tested under battlefield conditions and come up with much more firepower to compete with UltraSparc III servers above four-way platforms, one analyst said.

Santa Clara, Calif.-based Intel released a series of benchmarks recently that demonstrate Itanium 2's evolution in its McKinley version, which seems to demonstrate that it is far closer to what server builders are looking for in performance in a 64-bit processor than the chip's current and disappointing Merced version.

Performance improvements of 50 percent to 100 percent above the previous-generation chip are expected when the Itanium 2 McKinley makes its debut in Hewlett-Packard servers later this year, said Intel's Jason Waxman of the enterprise platform marketing division.

In enterprise resource planning applications, Intel's 1GHz I-2 chip is capable of twice the throughput of the original 800MHz Itanium, Intel said. In another benchmark comparison, Intel's I-2 was said to be twice as fast as a 1,050MHz UltraSparc III in floating point performance, according to SPECfp2000, published by the Standard Performance Evaluation Corp.

"With the Itanium 2 performance improvements, it looks like Intel has put more pressure on Sun earlier than we had anticipated," said Peter Kastner, executive VP of the Aberdeen Group in Boston.

"If you really want to know how the Itanium 2-based boxes will perform, wait until they are announced by the OEMs," Kastner added. "Clearly Intel wants to compare the Itanium with the best-in-breed of the RISC processors where Sun has the broadest market share."

Kastner said the benchmarking shown by Intel is impressive, as the Itanium 2 has trumped even Moore's Law in some performance tests. But the benchmarks that really matter will be those tested on SAP, PeopleSoft and Oracle software applications, and potential buyers should wait until those numbers emerge before making a purchasing decision.

A representative of Sun was quick to dispute the Intel claims, saying the tests don't really provide apples-to-apples comparisons across platforms. "To make real comparisons, you need to test the chips on real workloads," said Sue Kunz, director of marketing for Sun's Processor Products Group. Such real workloads include performance on PeopleSoft and Lotus Notes applications, she said.

"The power consumption numbers haven't really gone down," Kunz said. She even pointed to an HP box, which she said consisted predominantly of cooling equipment and fans.

The I-2 performance benchmarks are probably a little too close for comfort for Sun's liking, Kastner said. Intel is showing impressive numbers in one-way to four-way boxes. But Sun's UltraSparc chips can really strut their stuff in systems with up to 100 processors operating simultaneously, Kastner said. ■